Flat morphisms in connection with cohomology

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These are notes from my talk in the Summer 2026 coherent cohomology seminar in Münster. This is mostly based on section III.9 of Hartshorne's book. My thanks to Prof. Dr. Hartl for his guidance during my talk preparation, and to Connor Lane for helping me figure out the reduction in the proof of theorem 2.7.

1. Introduction: Recollections on flatness

Recall the notion of flatness from commutative algebra: an A-module (or A-algebra) M is flat if and only if the tensor product $- \otimes_A M$ is an exact functor. This is a very useful property to have, especially when working with (co)homological things, as (for example) exact functors commute with the formation of (co)homology on a chain complex. Additionally, we know that many common classes of modules/algebras such as free and projective modules as well as localizations are always flat, and we can even test flatness completely locally.

In the context of algebraic geometry, we have a notion of flatness which is relative: for $f\colon X \to Y$ a morphism of schemes and $\mathcal{F}$ an $\mathcal{O}_X$-module, we say $\mathcal{F}$ is flat over Y if and only if $\mathcal{F}_x$ is a flat $\mathcal{O}_{Y,f(x)}$-module for all $x \in X$. If $\mathcal{O}_X$ is flat over Y we simply say f is a flat morphism. This is maybe a bit of a mysterious condition geometrically, but it is a very nice algebraic condition akin to the one from classical commutative algebra. For instance, if $f = \mathrm{id}_X$, then $\mathcal{F}$ is flat over X if and only if the tensor product $- \otimes_{\mathcal{O}_X} \mathcal{F}\colon \mathcal{O}_X\text{-}\mathcal{M}\mathrm{od} \to \mathcal{O}_X\text{-}\mathcal{M}\mathrm{od}$ is an exact functor, and $f\colon X \to Y$ is a flat morphism if and only if the pullback $f^*\colon \mathcal{O}_Y\text{-}\mathcal{M}\mathrm{od} \to \mathcal{O}_X\text{-}\mathcal{M}\mathrm{od}$ is exact.

This suggests that at the very least, we should expect flat morphisms to behave well with cohomological operations on schemes and $\mathcal{O}_X$-modules. In this talk we will see a couple instances of this nice behavior with regards to two concepts introduced in the last talk, namely higher direct images and the Hilbert polynomial. In fact, we will see that in nice situations flatness is equivalent to a condition about Hilbert polynomials. We will also indicate how one can prove some more intrinsically geometric statements in the presence of flatness, essentially by reducing down to a situation where the problem is completely algebraic.

2. Flat base change

Note. *This section requires several (mildly tedious) technical lemmas. In the interest of having a clear exposition, we will defer the proofs of these to the appendix.*

One of the nice properties of flat morphisms arises in studying a very natural question about compatibility of inverse and (higher) direct images of sheaves. Consider a commutative diagram of schemes

$$(*) \qquad \begin{tikzcd} X' \arrow[r, "v"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "u"'] & Y \end{tikzcd}$$

Given a sheaf on one of these schemes, we can form its direct and/or inverse images in various ways, as summarized in the following diagrams:

$$\begin{tikzcd} \mathcal{O}_{X'}\text{-}\mathcal{M}od \arrow[r, "v_*"] \arrow[d, "g_*"'] & \mathcal{O}_X\text{-}\mathcal{M}od \arrow[d, "f_*"] \\ \mathcal{O}_{Y'}\text{-}\mathcal{M}od \arrow[r, "u_*"'] & \mathcal{O}_Y\text{-}\mathcal{M}od \end{tikzcd} \qquad \begin{tikzcd} \mathcal{O}_{X'}\text{-}\mathcal{M}od \arrow[d, "g_*"'] & \mathcal{O}_X\text{-}\mathcal{M}od \arrow[l, "v^*"'] \arrow[d, "f_*"] \\ \mathcal{O}_{Y'}\text{-}\mathcal{M}od & \mathcal{O}_Y\text{-}\mathcal{M}od \arrow[l, "u^*"] \end{tikzcd}$$

$$\begin{tikzcd} \mathcal{O}_{X'}\text{-}\mathcal{M}od \arrow[r, "v_*"] & \mathcal{O}_X\text{-}\mathcal{M}od \\ \mathcal{O}_{Y'}\text{-}\mathcal{M}od \arrow[r, "u_*"'] \arrow[u, "g^*"] & \mathcal{O}_Y\text{-}\mathcal{M}od \arrow[u, "f^*"'] \end{tikzcd} \qquad \begin{tikzcd} \mathcal{O}_{X'}\text{-}\mathcal{M}od & \mathcal{O}_X\text{-}\mathcal{M}od \arrow[l, "v^*"'] \\ \mathcal{O}_{Y'}\text{-}\mathcal{M}od \arrow[u, "g^*"] & \mathcal{O}_Y\text{-}\mathcal{M}od \arrow[l, "u^*"] \arrow[u, "f^*"'] \end{tikzcd}$$

It's very natural to ask whether these processes are always compatible (i.e. whether these diagrams commute in a suitable sense). In the case where we only take direct images or only take inverse images, it is easy to see that

$$f_* \circ v_*(\mathcal{F}) = (f \circ v)_*(\mathcal{F}) = (u \circ g)_*(\mathcal{F}) = u_* \circ g_*(\mathcal{F})$$

and

$$v^* \circ f^*(\mathcal{F}) \cong (f \circ v)^*(\mathcal{F}) = (u \circ g)^*(\mathcal{F}) \cong g^* \circ u^*(\mathcal{F}),$$

so we have a very nice (natural) compatibility in these cases. The other case, where we mix direct and inverse images, is a more subtle question. Asking the same question with higher direct images also introduces more difficulties. In general, there is always a natural morphism

$$u^* \circ R^i f_*(\mathcal{F}) \to R^i g_* \circ v^*(\mathcal{F}),$$

but this won't always be an isomorphism. For $i = 0$, there is an easy way to construct this morphism; note that by adjunction of u^* and u_*, it is equivalent to specify a morphism

$$f_*(\mathcal{F}) \to u_* \circ g_* \circ v^*(\mathcal{F}) = f_* \circ v_* \circ v^*(\mathcal{F}).$$

We see then that by functoriality of f_*, it suffices to have a morphism

$$\mathcal{F} \to v_* \circ v^*(\mathcal{F});$$

but for this we already have a natural morphism available – namely the unit of the adjunction $v^* \dashv v_*$. One could just as well do the same process using the adjunction $g^* \dashv g_*$ and the counit of $f^* \dashv f_*$ to obtain another morphism

$$u^* \circ f_*(\mathcal{F}) \to g_* \circ v^*(\mathcal{F}),$$

and it turns out these are the same. For the case with higher direct images, we can do a similar construction but we need to work a bit harder (following [Con]). We start again by noting that specifying a morphism

$$u^* \circ R^i f_*(\mathcal{F}) \to R^i g_* \circ v^*(\mathcal{F})$$

is equivalent to specifying a morphism

$$R^i f_*(\mathcal{F}) \to u_* \circ R^i g_* \circ v^*(\mathcal{F}).$$

Now recall [Har77, Prop. III.8.1] that we can describe the higher direct images $R^i f_*(\mathcal{F})$ as the sheafification of

$$U \mapsto H^i(f^{-1}(U), \mathcal{F}|_{f^{-1}(U)});$$

we will use this description to define our desired morphism.

Lemma 2.1 (Contravariance of sheaf cohomology [EGA, 0_{III}, 12.1.3, 12.1.4]). [1] *Let $\varphi \colon Z' \to Z$ be a morphism of schemes and $\mathcal{F}$ an $\mathcal{O}_Z$-module. For all $i \geq 0$ there exists a natural map of abelian groups*

$$H^i(Z, \mathcal{F}) \to H^i(Z', \varphi^*\mathcal{F}).$$

If Z is an A-scheme for some commutative ring A, then the above map is one of A-modules.

For $U \subset Y$ an open subset, we apply lemma 2.1 to the morphism $(v \circ f)^{-1}(U) \to f^{-1}(U)$ and the sheaf $\mathcal{F}|_{f^{-1}(U)}$. This yields for all $i \geq 0$ a natural morphism

$$H^i(f^{-1}(U), \mathcal{F}|_{f^{-1}(U)}) \to H^i((f \circ v)^{-1}(U), v^*\mathcal{F}|_{(f\circ v)^{-1}(U)}) = H^i(g^{-1}(u^{-1}(U)), v^*\mathcal{F}|_{g^{-1}(u^{-1}(U))}).$$

Applying the sheafification functor, we get a natural morphism

$$R^i f_*(\mathcal{F})(U) \to R^i g_*(v^*\mathcal{F})(u^{-1}(U))$$

which constitutes the morphism of sheaves $R^i f_*(\mathcal{F}) \to u_* \circ R^i g_* \circ v^*(\mathcal{F})$ we sought.

[1] In this citation style for EGA, "0_{III}" signifies the part of chapter 0 which was published in EGA III.

Remark 2.2. *This construction works more generally for any ringed space, which apart from schemes includes the interesting cases of abelian sheaves on any topological space and coherent sheaves on any smooth/complex manifold.*

Remark 2.3. *There are many different ways to construct the base change morphism found in the literature, particularly when one imposes more assumptions on the spaces/morphisms in the starting diagram and when one has more technology at one's disposal. While it seems to not be explicitly documented (and is not in general especially clear), it is implicit in the literature that seemingly any natural map* $u^* \circ R^i f_*(\mathcal{F}) \to R^i g_* \circ v^*(\mathcal{F})$ *that one constructs will be the same as the morphism that we have defined.*

We should expect that these base change morphisms behave well with respect to operations such as pasting squares. Indeed, this is the case:

Lemma 2.4. *Suppose we have a commutative diagram of schemes*

$$\begin{tikzcd} X'' \arrow[r, "w"] \arrow[d, "h"'] & X' \arrow[r, "v"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ Y'' \arrow[r, "t"'] & Y' \arrow[r, "u"'] & Y \end{tikzcd} \tag{\#}$$

and $\mathcal{F}$ *an* $\mathcal{O}_X$*-module. Then the triangle*

$$\begin{tikzcd} t^* \circ u^* \circ R^i f_*(\mathcal{F}) \arrow[r] \arrow[dr] & t^* \circ R^i g_* \circ v^*(\mathcal{F}) \arrow[d] \\ & R^i h_* \circ w^* \circ v^*(\mathcal{F}) \end{tikzcd} \tag{!}$$

where the vertical arrow is the base change morphism on the left square for the $\mathcal{O}_{X'}$*-module* $v^*\mathcal{F}$*, the horizontal arrow is* t^* *applied to the base change morphism on the right square for* $\mathcal{F}$*, and the diagonal arrow is the base change morphism on the outer rectangle for* $\mathcal{F}$*, commutes.*

Now that we have the base change morphism $u^* \circ R^i f_*(\mathcal{F}) \to R^i g_* \circ v^*(\mathcal{F})$, we may ask when this is an isomorphism. The crucial assumption that we always need to enforce for this to hold is that the square of schemes ($*$) we started with is Cartesian[2] (i.e. defines a pullback). Under this assumption there are a couple of easy cases where the base change is an isomorphism:

Proposition 2.5. *Suppose the square* ($*$) *is Cartesian and the* $\mathcal{O}_X$*-module* $\mathcal{F}$ *is quasicoherent; then the base change map is an isomorphism in the following two situations:*

(a) All schemes in the square are affine;
(b) The morphism $f \colon X \to Y$ *is affine.*

[2]and maybe rightly the base change morphism should only be called "base change" when the square is, well, a base change

Proof. Left as an exercise to the reader. The key idea is to deduce (b) from (a) and reduce (a) to the associativity of the tensor product.[3]

The following is a special case of theorem 2.7, but we will need it in the proof of that theorem so we state it separately.

Lemma 2.6. *Suppose the square* ($*$) *is Cartesian and* u *is an open immersion. Then the base change morphism is an isomorphism.*

Proof. Clear by tracking through the construction and noticing that cohomology and higher direct images are both local [Har77, Cor. III.8.2].

We now come to the main theorem of this section: the base change morphism is an isomorphism for quasicoherent sheaves when the relevant pullbacks are exact (i.e. when the corresponding morphisms are flat).

Theorem 2.7 (Flat base change [Har77, Prop. III.9.3])**.** *Let* $f\colon X \to Y$ *be a separated morphism of Noetherian schemes, and let* $\mathcal{F}$ *be a quasicoherent sheaf on* X*. Let* $u\colon Y' \to Y$ *be a flat morphism of Noetherian schemes, and suppose we have morphisms* $v\colon X' \to X$ *and* $g\colon X' \to Y'$ *from a Noetherian scheme* X' *making the following a Cartesian diagram:*

$$\begin{tikzcd} X' \arrow[r, "v"] \arrow[d, "g"'] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "f"] \\ Y' \arrow[r, "u"'] & Y \end{tikzcd}$$

Then

(a) for all $i \geq 0$*, the base change morphism*

$$u^* R^i f_*(\mathcal{F}) \to R^i g_*(v^* \mathcal{F})$$

is an isomorphism;

(b) if $Y' = \operatorname{Spec} A'$ *and* $Y = \operatorname{Spec} A$ *are affine, then the natural morphisms*

$$H^i(X, \mathcal{F}) \otimes_A A' \xrightarrow{\sim} H^i(X', v^* \mathcal{F})$$

obtained by tensoring the morphism from lemma 2.1 with A' *are isomorphisms for all* $i \geq 0$*.*

Proof. We first show that for (a) we may assume both Y' and Y are affine. First, if we know the statement when both are affine, then we can deduce that the statement also holds whenever Y is affine: for affine open $j\colon U \hookrightarrow Y'$ we form the pullback rectangle

$$\begin{tikzcd} U_{X'} \arrow[r, hook] \arrow[d] & X' \arrow[r] \arrow[d] & X \arrow[d] \\ U \arrow[r, hook] & Y' \arrow[r] & Y \end{tikzcd}$$

[3] One should also really check that in the affine situation, the base change morphism agrees with the isomorphism encoding associativity of the tensor product!

and note that we know that the base change map is an isomorphism for

- the outer rectangle, by assumption
- and the left square, by lemma 2.6.

But then lemma 2.4 implies j^* applied to the base change morphism for the right square is an isomorphism, and since this holds for any affine open $U \subset Y'$, the same is true for the base change morphism itself. We make a slightly more involved version of this argument to show that if we know the statement is true whenever Y is affine, then we can deduce that it holds in general. For affine open $j\colon U \hookrightarrow Y$, we form the pullback of the entire diagram

$$\begin{tikzcd}
& X'_U \arrow[rr] \arrow[dl, hook] \arrow[dd] & & X_U \arrow[dl, hook] \arrow[dd] \\
X' \arrow[rr, crossing over] \arrow[dd] & & X & \\
& U' \arrow[rr] \arrow[dl, hook] & & U \arrow[dl, hook] \\
Y' \arrow[rr] & & Y \arrow[from=uu, crossing over] &
\end{tikzcd}$$

and note that by assumption we know the theorem holds for the back face. We also know that it holds for the left and right faces by lemma 2.6. This in particular implies that the theorem holds for the composite of the back and right faces per lemma 2.4, but by commutativity of the top and bottom faces, this is precisely the same as the composite of the left and front faces. Now we again deduce that the base change map for the front face is an isomorphism after pulling back to U', and as U runs over an affine open over of Y, U' will constitute an open cover of Y' and we can conclude that the base change map is an isomorphism.

Now assuming $Y = \operatorname{Spec} A$ and $Y' = \operatorname{Spec} A'$ are affine, by [Har77, Prop. III.8.5], we know that

$$R^i f_*(\mathcal{F}) \cong \widetilde{H^i(X, \mathcal{F})} \quad \text{and} \quad R^i g_*(v^*\mathcal{F}) \cong \widetilde{H^i(X', v^*\mathcal{F})}.$$

Combining this with locality [Har77, Cor. III.8.2], we see that the sheafification process in the construction of the base change morphism does nothing. Hence, the A-module map corresponding to the morphism of quasicoherent $\mathcal{O}_Y$-modules $R^i f_*\mathcal{F} \to u_* \circ R^i g_* \circ v^*(\mathcal{F})$ is precisely the morphism in lemma 2.1. On modules, the adjunction $u^* \dashv u_*$ is just tensor-Hom, so we obtain a map

$$H^i(X, \mathcal{F}) \otimes_A A' \to H^i(X', v^*\mathcal{F});$$

to establish (a) it suffices to prove that this is an isomorphism, and this is also exactly the claim in (b).

As X is separated over an affine base, any finite intersection of affine open subsets is again affine [Har77, Ex. II.4.3], hence by [Har77, Thm. III.4.5] we can compute $H^i(X, \mathcal{F})$ by Čech cohomology with respect to an affine open cover $\mathcal{U}$ of X. Moreover, as u is affine, its pullback ν is also affine, and hence $\mathcal{U}' := \{\nu^{-1}(U) : U \in \mathcal{U}\}$ is an affine open cover of X'. g is separated as it is the pullback of f, hence we may also compute $H^i(X', \nu^*\mathcal{F})$ by Čech cohomology with respect to the cover $\mathcal{U}'$. But the Čech complex in this case is

$$0 \longrightarrow \prod_{i_0} \nu^*\mathcal{F}(\nu^{-1}(U_{i_0})) \longrightarrow \prod_{i_0<i_1} \nu^*\mathcal{F}(\nu^{-1}(U_{i_0,i_1})) \longrightarrow \cdots$$

As X is Noetherian and in particular quasi-compact, we may choose $\mathcal{U}$ to be finite so that all products in the Čech complexes $C^\bullet(\mathcal{U}, \mathcal{F})$ and $C^\bullet(\mathcal{U}', \nu^*\mathcal{F})$ are finite and coincide with direct sums. Now we claim that $\nu^*\mathcal{F}(\nu^{-1}(U)) \simeq \mathcal{F}(U) \otimes_A A'$ for any affine open subscheme $U \subset X$. Indeed, we may check this after replacing $\mathcal{F}$ with $\mathcal{F}|_U$ and $\nu^*\mathcal{F}$ with $\nu^*\mathcal{F}|_{\nu^{-1}(U)}$, where we know that $\mathcal{F}|_U = \widetilde{\mathcal{F}(U)}$ for the $\mathcal{O}_X(U)$-module $\mathcal{F}(U)$. As $\nu^{-1}(U)$ is affine, the pullback coincides with the tensor product i.e.

$$\nu^*\mathcal{F}|_{\nu^{-1}(U)} = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \widetilde{\mathcal{O}_{X'}}(\nu^{-1}(U)).$$

But we can also check via pullback pasting that $\nu^{-1}(U) = Y' \times_Y U$, or since these are affine,

$$\mathcal{O}_{X'}(\nu^{-1}(U)) = A' \otimes_A \mathcal{O}_X(U).$$

By associativity of the tensor product, we conclude that

$$\nu^*\mathcal{F}|_{\nu^{-1}(U)}(\nu^{-1}(U)) = \mathcal{F}(U) \otimes_A A'$$

as claimed. Substituting this into the Čech complex $C^\bullet(\mathcal{U}', \nu^*\mathcal{F})$, we have

$$0 \longrightarrow \bigoplus_{i_0} \mathcal{F}(U_{i_0}) \otimes_A A' \longrightarrow \bigoplus_{i_0<i_1} \mathcal{F}(U_{i_0,i_1}) \otimes_A A' \longrightarrow \cdots$$

Since the tensor product commutes with direct sums, we find that

$$C^\bullet(\mathcal{U}', \nu^*\mathcal{F}) \cong C^\bullet(\mathcal{U}, \mathcal{F}) \otimes_A A'.$$

The assumption that f is flat means precisely that $- \otimes_A A'$ is exact, so the tensor product also commutes with homology. Thus we obtain an isomorphism

$$\check{H}^i(\mathcal{U}', \nu^*\mathcal{F}) \cong \check{H}^i(\mathcal{U}, \mathcal{F}) \otimes_A A'$$

on Čech cohomology. One needs to check that this is the same as the base change morphism when we pass between sheaf and Čech cohomology, and this is recorded in lemma 2.8

Lemma 2.8 ([EGA, 0_{III}, 12.1.4])**.** *Let $\varphi : Z' \to Z$ be a morphism of schemes and $\mathcal{F}$ an $\mathcal{O}_Z$-module. For any open cover $\mathcal{U}$ of Z, let $\mathcal{U}' := \{\varphi^{-1}(U) : U \in \mathcal{U}\}$ be the pullback of $\mathcal{U}$ to Z'. Then the natural square*

$$\begin{tikzcd} \check{H}^i(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d] & \check{H}^i(\mathcal{U}', \varphi^*\mathcal{F}) \arrow[d] \\ H^i(Z, \mathcal{F}) \arrow[r] & H^i(Z', \varphi^*\mathcal{F}) \end{tikzcd}$$

commutes for all $i \geq 0$.

Remark 2.9. *This proof works more generally without any Noetherian assumptions, as long as* f *is a quasicompact morphism. More generally, the theorem is still true as long as* f *is quasicompact and* quasiseparated *(with* u *still flat and no further assumptions on the schemes), but this is more difficult.*

We obtain the following consequence of theorem 2.7, which we need later for the proof of theorem 3.2.

Proposition 2.10. *Let* A *be a Noetherian local domain,* X *a projective* A*-scheme, and* $\mathcal{F}$ *a coherent sheaf on* X*. Then for all* $t \in T = \operatorname{Spec} A$ *and* $m \gg 0$ *we have*

$$H^0(X_t, \mathcal{F}_t(m)) \cong H^0(X, \mathcal{F}(m)) \otimes_A k(t).$$

Proof. Set $T' := \operatorname{Spec} \mathcal{O}_{T,t}$ and recall that $T' \to T$ is flat as a localization. As cohomology over an affine base commutes with affine flat base change (theorem 2.7(b)), we reduce to the case where $t \in T$ is the closed point. Take a presentation of $k(t)$ as an A-module

$$A^q \longrightarrow A \longrightarrow k(t) \longrightarrow 0, \tag{1}$$

and consider the corresponding exact sequence of sheaves on X

$$\mathcal{F}^q \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_t \longrightarrow 0.$$

By [Har77, Ex. III.5.10] we have an exact sequence

$$H^0(X, \mathcal{F}(m))^q \longrightarrow H^0(X, \mathcal{F}(m)) \longrightarrow H^0(X_t, \mathcal{F}_t(m)) \longrightarrow 0$$

for $m \gg 0$ (twist enough until all higher cohomologies vanish). We can also tensor the original exact sequence (1) with $H^0(X, \mathcal{F}(m))$, and we get a commutative diagram with exact rows

$$\begin{tikzcd}
H^0(X, \mathcal{F}(m)) \otimes_A A^q \arrow[r] \arrow[d] & H^0(X, \mathcal{F}(m)) \otimes_A A \arrow[r] \arrow[d] & H^0(X, \mathcal{F}(m)) \otimes_A k(t) \arrow[r] \arrow[d] & 0 \\
H^0(X, \mathcal{F}(m))^q \arrow[r] & H^0(X, \mathcal{F}(m)) \arrow[r] & H^0(X_t, \mathcal{F}_t(m)) \arrow[r] & 0,
\end{tikzcd}$$

where we know that the left two vertical maps are isomorphisms since A and A^q are flat A-modules (theorem 2.7(b)). The snake lemma shows that the right vertical map is the claimed isomorphism.

3. Flatness and the Hilbert polynomial

In this section we prove theorem 3.2, which lets us detect flatness by computing Hilbert polynomials. Most of the work has already been done in the previous section, between the flat base change theorem (particularly part (b)) and proposition 2.10.

Lemma 3.1 ([Har77, Lemma II.8.9]). *Let A be a Noetherian local domain with residue field k and field of fractions K, and let M be a finitely generated A-module. Then M is free if and only if $\dim_k M \otimes_A k = \dim_K M \otimes_A K$, and in this case the dimensions over k and K coincide with $\operatorname{rank}_A M$.*

Proof. If $M \cong A^r$ is free, we clearly have $M \otimes_A k \cong k^r$ and $M \otimes_A K \cong K^r$, and the assertions are clear.

In the converse direction, let $r \coloneqq \dim_k M \otimes_A k = \dim_K M \otimes_A K$. Nakayama's lemma allows us to lift generators of $M \otimes_A k$ and see that M is generated as an A-module by r elements. In particular, this means we have a surjection $A^r \twoheadrightarrow M$; letting N be the kernel of this surjection, we have an exact sequence

$$0 \longrightarrow N \longrightarrow A^r \longrightarrow M \longrightarrow 0.$$

We wish to show that $N = 0$; we observe that since N is a submodule of A^r and A is a domain, N is torsion-free. Hence it suffices to show that N is also a torsion A-module. The easiest way to do this is to check that $N \otimes_A K = 0$. As K is flat over A (since it's a localization of A), tensoring preserves exactness:

$$0 \longrightarrow N \otimes_A K \longrightarrow K^r \longrightarrow M \otimes_A K \longrightarrow 0.$$

As $M \otimes_A K$ has dimension r, the surjection $K^r \twoheadrightarrow M \otimes_A K$ is in fact an isomorphism, so $N \otimes_A K = 0$ as desired.

Theorem 3.2 ([Har77, Thm. III.9.9]). *Let T be an integral Noetherian scheme, $f\colon X \to T$ a projective T-scheme, and $\mathcal{F}$ a coherent sheaf on X. For each point $t \in T$, let $P_{\mathcal{F}_t} \in \mathbf{Q}[z]$ be the Hilbert polynomial of $\mathcal{F}$ on the fibre X_t, considered as a projective $k(t)$-scheme. Then the following are equivalent:*

- *(a) $\mathcal{F}$ is flat over T;*
- *(b) $f_*\mathcal{F}(m)$ is a finite locally free sheaf (vector bundle) for $m \gg 0$;*
- *(c) the polynomial $P_{\mathcal{F}_t}$ is independent of t.*

Proof. Firstly, this assertion is purely local on T, so we may assume without loss of generality that $T = \operatorname{Spec} A$ is affine. Moreover, we claim that we can test all of these conditions after passing to localizations. For (a) this is by definition of flatness; for (b) this is via Nakayama's lemma since $f_*\mathcal{F}(m)$ is a finitely generated A-module [Har77, Thm. III.5.2.(a)]. For (c), we may certainly compute $P_{\mathcal{F}_t}$ after localizing at t; however, we can also compute $P_{\mathcal{F}_s}$ on $\operatorname{Spec} \mathcal{O}_{T,t}$ for any generization s of t. Since T is integral, we can compute the Hilbert polynomial at the generic point in any localization, hence (c) can also be checked after passing to localizations. Thus we may assume $T = \operatorname{Spec} A$ for A an integral Noetherian local ring. We note that in this case, (b) is equivalent to $\Gamma(T, f_*\mathcal{F}(m)) = H^0(X, \mathcal{F}(m))$ being a finite free A-module. We will prove (a) $\iff$ (b), then (b) $\iff$ (c).

(a) $\Rightarrow$ (b): First assume $\mathcal{F}$ is flat over T; as X is projective over T it is separated over T, and hence we may compute $H^i(X, \mathcal{F}(m)) = \Gamma(T, R^i f_*\mathcal{F}(m))$ by Čech cohomology of an affine open cover $\mathcal{U}$. As $\mathcal{F}$ is flat, $\mathcal{F}(U)$ is a flat A-module for $U \subset X$ open, and hence each term of the Čech complex $C^\bullet(\mathcal{U}, \mathcal{F}(m))$ is a flat A-module. However, we know that the cohomology of $\mathcal{F}(m)$ vanishes for $m \gg 0$ [Har77, Thm. III.5.2.(b)], so the Čech complex is exact. In particular, we have an exact sequence of A-modules

$$0 \longrightarrow H^0(X, \mathcal{F}(m)) \longrightarrow C^0 \longrightarrow C^1 \longrightarrow \cdots$$

where each C^i is flat. By splitting this into short exact sequences, we conclude that $H^0(X, \mathcal{F}(m))$ is flat as well. However, we also have that $H^0(X, \mathcal{F}(m))$ is finitely generated over A [Har77, Thm. III.5.2.(a)], hence projective.

(b) $\implies$ (a) Assume now that $H^0(X, \mathcal{F}(m))$ is finite free for $m \gg 0$. We can assume that $X = \mathbf{P}^n_A$: indeed, if $j\colon X \hookrightarrow \mathbf{P}^n_A$ is a closed immersion factoring f as follows

$$\begin{tikzcd} X \arrow[r, hook, "j"] \arrow[dr, "f"'] & \mathbf{P}^n_A \arrow[d, "g"] \\ & T, \end{tikzcd}$$

we have that $f_*\mathcal{F}(m) = g_* \circ j_*\mathcal{F}(m)$ for any m. Moreover, flatness is checked at stalks and the stalks of $j_*\mathcal{F}$ are

$$(j_*\mathcal{F})_z = \begin{cases} \mathcal{F}_x & \text{if } z = j(x) \\ 0 & \text{else,} \end{cases}$$

so $\mathcal{F}$ is flat over T if and only if $j_*\mathcal{F}$ is flat over T. Hence we replace X with $\mathbf{P}^n_A = \operatorname{Proj} S$ and $\mathcal{F}$ with $j_*\mathcal{F}$. Let m_0 be such that $H^0(X, \mathcal{F}(m))$ is free for all $m \geq m_0$, and let M be the graded S-module

$$M := \bigoplus_{m \geq m_0} H^0(X, \mathcal{F}(m)).$$

This agrees with

$$\Gamma_*(\mathcal{F}) := \bigoplus_{m \in \mathbf{Z}} H^0(X, \mathcal{F}(m))$$

in degrees $m \geq m_0$, so by [Har77, Prop. II.5.15] we have that $\mathcal{F} = \widetilde{\Gamma_*\mathcal{F}} = \tilde{M}$. As M is free (and hence flat) over A, it follows that $\mathcal{F}$ is flat over A as well, since each of its stalks is a direct summand of a localization of M [Har77, Prop. II.5.11.(a)], hence itself flat.

(b) $\implies$ (c) Now we assume again that $H^0(X, \mathcal{F}(m))$ is finite free for $m \gg 0$, and we will prove that the Hilbert polynomial $P_{\mathcal{F}_t}$ is independent of t. Recall that since $\mathcal{F}_t(m)$ has vanishing cohomology for $m \gg 0$, we have

$$P_{\mathcal{F}_t}(m) = \dim_k H^0(X_t, \mathcal{F}_t(m))$$

for $m \gg 0$. By proposition 2.10, we have (for $m \gg 0$)

$$H^0(X, \mathcal{F}(m)) \otimes_A k(t) \cong H^0(X_t, \mathcal{F}_t(m))$$

for all $t \in T$. As $H^0(X, \mathcal{F}(m))$ is free for large m, we see that the dimension of the left hand side is equal to $\operatorname{rank}_A H^0(X, \mathcal{F}(m))$, which is independent of t.

(c) $\implies$ (b) Via lemma 3.1, it is enough to check that

$$\dim_k H^0(X, \mathcal{F}(m)) \otimes_A k = \dim_K H^0(X, \mathcal{F}(m)) \otimes_A K$$

for k the residue field and K the fraction field of A. Again we have

$$H^0(X, \mathcal{F}(m)) \otimes_A k(t) \cong H^0(X_t, \mathcal{F}_t(m))$$

per proposition 2.10, and by assumptions the dimensions of the right side are equal for t the special and generic points.

Remark 3.3. *This theorem can be generalized to the case where X is proper of finite presentation over T. One can remove the integrality assumption on T in favor of a reducedness assumption, if (c) is modified to "$P_{\mathcal{F}_t}$ depends only on the irreducible component of t." More precisely, the implications (a) $\iff$ (b) $\implies$ (c) are true with no assumption on T (even Noetherianness can be discarded with appropriate care), but the implication (c) $\implies$ (b) requires T to be reduced and locally Noetherian. See [GW23, (23.29)] for more.*

4. Flat families of schemes

In this section we prove a theorem which tells us that flatness ensures that dimensions behave as we should expect in fibres. Unlike the previous two theorems, this will not involve any cohomology; rather it will involve a series of reductions to a commutative algebra statement, for which we can apply flatness nicely. Much of the proof can be stated more geometrically than has been done here (Hartshorne does this, for example), but many of the key reductions have to be argued on the commutative algebra side, so we opt to just stay in that realm.

For any scheme S and any point $s \in S$, we let $\dim_s S$ denote the Krull dimension of the local ring $\mathcal{O}_{S,s}$.

Theorem 4.1 ([Har77, Prop. III.9.5])**.** *Let k be a field and let $f \colon X \to Y$ be a flat morphism of finite-type k-schemes. For any point $x \in X$, we have*

$$\dim_x(X_{f(x)}) = \dim_x X - \dim_{f(x)} Y.$$

Proof. First, the statement is local: if $U \subset Y$ is an affine open containing $f(x)$, then x lifts to $U \times_Y X$. Clearly $\dim_{f(x)} Y$ can be computed in U, and pasting of pullbacks also shows that $\dim_x(X_{f(x)})$ can be computed after replacing X with $X \times_Y U$. Moreover, $X \times_Y U \to X$ is an open immersion, so we also have $\dim_x X = \dim_x(X \times_Y U)$. Assuming now that Y is affine, we can again take an open $V \subset X$ containing x and see that for similar reasons we have $\dim_x X = \dim_x V$ and that we can compute $\dim_x(X_{f(x)})$ after base changing to V.

Now we can assume that both $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are affine, hence the statement reduces to the following: for $f\colon B \to A$ a flat morphism of finitely generated k-algebras and any prime ideal $\mathfrak{p} \subset A$, we have

$$\dim(A \otimes_B \kappa(f^{-1}\mathfrak{p}))_{\mathfrak{p}} = \dim A_{\mathfrak{p}} - \dim B_{f^{-1}\mathfrak{p}}.$$

We can reduce this to the case where B is local; certainly, we can compute both $\dim B_{f^{-1}\mathfrak{p}}$ and $A \otimes_B \kappa(f^{-1}\mathfrak{p})$ after localizing at $f^{-1}\mathfrak{p}$. Moreover, we have isomorphisms

$$(A \otimes_B B_{f^{-1}\mathfrak{p}}) \otimes_A A_{\mathfrak{p}} \cong B_{f^{-1}\mathfrak{p}} \otimes_B A_{\mathfrak{p}} \cong A_{\mathfrak{p}},$$

so $\dim A_{\mathfrak{p}}$ can also be computed after localization at $f^{-1}\mathfrak{p}$.

The final reduction we make is to assume that B is reduced. For this, we note that $B \to B_{\mathrm{red}}$ is an integral morphism which is surjective after applying Spec, and by [Sta25, Tag 00OH] plus the going-up theorem, any such morphism preserves dimension. Moreover, the quotient $B \to \kappa(f^{-1}\mathfrak{p})$ clearly factors through B_{red}, so the only thing we need to check is that $\dim A_{\mathfrak{p}} = \dim(B_{\mathrm{red}} \otimes_B A)_{\mathfrak{p}}$. It is clear that the morphism $A_{\mathfrak{p}} \to B_{\mathrm{red}} \otimes_B A_{\mathfrak{p}}$ is integral, as the base change of an integral morphism, so it suffices to check that it is surjective after applying Spec. This amounts to showing that any morphism $A_{\mathfrak{p}} \to L$ to a field L must factor through $B_{\mathrm{red}} \otimes_B A_{\mathfrak{p}}$, and for that we consult the following diagram:

$$\begin{tikzcd}
& & L \\
B_{\mathrm{red}} \arrow[r] & B_{\mathrm{red}} \otimes_B A_{\mathfrak{p}} & \\
B \arrow[u] \arrow[r] & A_{\mathfrak{p}} \arrow[u] \arrow[uur] &
\end{tikzcd}$$

Since L is reduced, the map $B \to A_{\mathfrak{p}} \to L$ factors through B_{red}, and now since the tensor product is the pushout in $\mathcal{CR}ing$ we obtain the desired factorization through $B_{\mathrm{red}} \otimes_B A_{\mathfrak{p}}$.

Summarizing, we have reduced to showing that for $f\colon B \to A$ a flat morphism of finite-type k-algebras, where B is reduced and local with maximal ideal $\mathfrak{m}$, and $\mathfrak{p} \subset A$ a prime ideal such that $f(\mathfrak{m}) \subset \mathfrak{p}$, we have

$$\dim(A \otimes_B \kappa(\mathfrak{m}))_{\mathfrak{p}} = \dim A_{\mathfrak{p}} - \dim B.$$

We show this by inducting on $\dim B$. The case $\dim B = 0$ means $B = \kappa(\mathfrak{m})$ is a field, so the result is obvious. Now for $\dim B = n + 1$ (assuming the theorem true for $\dim B = n$), we can find some $b \in \mathfrak{m}$ which is a non-zero-divisor. Since f is flat, $f(b)$ is also a non-zero-divisor in A; now Krulls Hauptidealsatz shows that

$$\dim(B/bB) = \dim B - 1 \quad \text{and} \quad \dim(A/f(b)A)_{\mathfrak{p}} = \dim(A_{\mathfrak{p}}/f(b)A_{\mathfrak{p}}) = \dim A_{\mathfrak{p}} - 1.$$

Moreover, (as usual) we can compute the fibre after doing this base extension, so it suffices to prove the claim for B/bB and $A/f(b)A$. But these are covered by the induction hypothesis (after possibly passing to $(B/bB)_{\mathrm{red}}$) so we are done.

We now prove the lemmas we postponed earlier.

Lemma 2.1 (Contravariance of sheaf cohomology [EGA, 0_{III}, 12.1.3, 12.1.4]). [4] *Let $\varphi\colon Z' \to Z$ be a morphism of schemes and $\mathcal{F}$ an $\mathcal{O}_Z$-module. For all $i \geq 0$ there exists a natural map of abelian groups*

$$H^i(Z, \mathcal{F}) \to H^i(Z', \varphi^*\mathcal{F}).$$

If Z is an A-scheme for some commutative ring A, then the above map is one of A-modules.

Proof. First, let $\varphi^{-1}\colon \mathcal{A}b(Z') \to \mathcal{A}b(Z)$ denote the pullback of abelian sheaves. We recall the definition of the pullback of modules:

$$\varphi^*\mathcal{F} := \mathcal{O}_{Z'} \otimes_{\varphi^{-1}\mathcal{O}_Z} \varphi^{-1}\mathcal{F}.$$

Hence we see by functoriality that there is already a natural map

$$H^i(Z', \varphi^{-1}\mathcal{F}) \to H^i(Z', \varphi^*\mathcal{F}),$$

and so it suffices to construct a natural morphism

$$H^i(Z, \mathcal{F}) \to H^i(Z', \varphi^{-1}\mathcal{F}).$$

For this, choose an injective resolution $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ in $\mathcal{A}b(Z)$ and an injective resolution $0 \to \varphi^{-1}\mathcal{F} \to \mathcal{J}^\bullet$ in $\mathcal{A}b(Z')$. Since the pullback in abelian sheaves is exact, we have that

$$0 \longrightarrow \varphi^{-1}\mathcal{F} \longrightarrow \varphi^{-1}\mathcal{I}^\bullet$$

is still exact. The proof that morphisms of abelian sheaves $\mathcal{F} \to \mathcal{G}$ extend to chain morphisms between their injective resolutions requires only that the resolution of $\mathcal{F}$ is exact (and not that it is injective). Hence the identity map $\varphi^{-1}\mathcal{F} \to \varphi^{-1}\mathcal{F}$ extends to a morphism of chain complexes $\varphi^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$. Moreover, when we apply $\Gamma(Z', -)$ and take cohomology, the induced maps

$$H^i(\Gamma(Z', \varphi^{-1}\mathcal{I}^\bullet)) \to H^i(\Gamma(Z', \mathcal{J}^\bullet)) = H^i(Z, \varphi^{-1}\mathcal{F})$$

are independent of the choice of morphism $\varphi^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$. But now we note that there is a natural morphism $\Gamma(Z, \mathcal{I}^\bullet) \to \Gamma(Z', \varphi^{-1}\mathcal{I}^\bullet)$ by definition of the inverse image sheaf (this also comes from the unit of the adjunction $\varphi^{-1} \dashv \varphi_*$), and hence the composition

$$H^i(Z, \mathcal{F}) = H^i(\Gamma(Z, \mathcal{I}^\bullet)) \to H^i(\Gamma(Z', \varphi^{-1}\mathcal{I}^\bullet)) \to H^i(\Gamma(Z', \mathcal{J}^\bullet)) = H^i(Z, \varphi^{-1}\mathcal{F})$$

is the map we were after.

If Z is an A-scheme, it is not immediately clear that the maps we have constructed are A-module morphisms, since we explicitly passed through $H^i(Z, \varphi^{-1}\mathcal{F})$, which is just an abelian group. To argue that these are indeed A-module maps we adapt a clever argument from EGA. We recall that by definition, we have a map of sheaves of rings $\mathcal{O}_Z \to \mathcal{O}_{Z'}$ and a ring homomorphism $A \to \mathcal{O}_Z(Z)$; hence we can see via restrictions that every $a \in A$

[4] In this citation style for EGA, "0_{III}" signifies the part of chapter 0 which was published in EGA III.

defines an endomorphism $a\colon \mathcal{F} \to \mathcal{F}$, which is given by multiplication by a on sections. Similarly, we get an endomorphism $a'\colon \varphi^*\mathcal{F} \to \varphi^*\mathcal{F}$ via the composite $A \to \mathcal{O}_Z \to \mathcal{O}_{Z'}$, and moreover since the morphism $\mathcal{O}_Z \to \mathcal{O}_{Z'}$ is one of sheaves, we find that the square

$$\begin{tikzcd} H^0(Z,\mathcal{F}) \arrow[r] \arrow[d, "a"'] & H^0(Z',\varphi^*\mathcal{F}) \arrow[d, "a'"] \\ H^0(Z,\mathcal{F}) \arrow[r] & H^0(Z',\varphi^*\mathcal{F}) \end{tikzcd}$$

must commute. Since this diagram commutes for H^0 and the induced maps on higher cohomology (from the chain-level arguments and functoriality) are uniquely determined by the induced maps on H^0, we find that for all $i \geq 0$ the square

$$\begin{tikzcd} H^i(Z,\mathcal{F}) \arrow[r] \arrow[d, "a"'] & H^i(Z',\varphi^*\mathcal{F}) \arrow[d, "a'"] \\ H^i(Z,\mathcal{F}) \arrow[r] & H^i(Z',\varphi^*\mathcal{F}) \end{tikzcd}$$

is forced to commute. But this commutativity for all $a \in A$ is exactly the statement that the maps $H^i(Z,\mathcal{F}) \to H^i(Z',\varphi^*\mathcal{F})$ are A-module maps! 

Lemma 2.4. *Suppose we have a commutative diagram of schemes*

$$(\#) \qquad \begin{tikzcd} X'' \arrow[r, "w"] \arrow[d, "h"'] & X' \arrow[r, "v"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ Y'' \arrow[r, "t"'] & Y' \arrow[r, "u"'] & Y \end{tikzcd}$$

and $\mathcal{F}$ an $\mathcal{O}_X$-module. Then the triangle

$$(!) \qquad \begin{tikzcd} t^* \circ u^* \circ R^i f_*(\mathcal{F}) \arrow[r] \arrow[dr] & t^* \circ R^i g_* \circ v^*(\mathcal{F}) \arrow[d] \\ & R^i h_* \circ w^* \circ v^*(\mathcal{F}) \end{tikzcd}$$

where the vertical arrow is the base change morphism on the left square for the $\mathcal{O}_{X'}$-module $v^*\mathcal{F}$, the horizontal arrow is t^* applied to the base change morphism on the right square for $\mathcal{F}$, and the diagonal arrow is the base change morphism on the outer rectangle for $\mathcal{F}$, commutes.

Proof. First consider the triangle

$$(!!) \qquad \begin{tikzcd} u^* \circ R^i f_*(\mathcal{F}) \arrow[r] \arrow[dr] & R^i g_* \circ v^*(\mathcal{F}) \arrow[d] \\ & t_* \circ R^i h_* \circ w^* \circ v^*(\mathcal{F}) \end{tikzcd}$$

where the horizontal map is the base change for the right square in (#) and the vertical and diagonal maps are the ones obtained from the vertical and diagonal maps in (!) via the adjunction $t^* \dashv t_*$. The naturality of the adjunction data means that the triangle (!!)

commutes if and only if (!) commutes. In a similar manner, these commute if and only if the triangle obtained from (!!) by adjunction on the horizontal and diagonal arrows and u_* on the vertical arrow

$$\text{(!!!)} \qquad \begin{tikzcd} R^i f_*(\mathcal{F}) \arrow[r] \arrow[dr] & u_* \circ R^i g_* \circ v^*(\mathcal{F}) \arrow[d] \\ & u_* \circ t_* \circ R^i h_* \circ w^* \circ v^*(\mathcal{F}) \end{tikzcd}$$

commutes. Now the claim is a matter of tracking through the definitions of these maps and seeing that they must agree.

Lemma 2.8 ([EGA, 0_{III}, 12.1.4])**.** *Let $\varphi\colon Z' \to Z$ be a morphism of schemes and $\mathcal{F}$ an $\mathcal{O}_Z$-module. For any open cover $\mathcal{U}$ of Z, let $\mathcal{U}' := \{\varphi^{-1}(U)\colon U \in \mathcal{U}\}$ be the pullback of $\mathcal{U}$ to Z'. Then the natural square*

$$\begin{tikzcd} \check{H}^i(\mathcal{U},\mathcal{F}) \arrow[r] \arrow[d] & \check{H}^i(\mathcal{U}',\varphi^*\mathcal{F}) \arrow[d] \\ H^i(Z,\mathcal{F}) \arrow[r] & H^i(Z',\varphi^*\mathcal{F}) \end{tikzcd}$$

commutes for all $i \geq 0$.

Proof. We first recall the definition of the maps in the square. The bottom map is the map from lemma 2.1. For the top map, we assemble the canonical maps

$$H^0(U,\mathcal{F}) \to H^0(f^{-1}(U),\varphi^*\mathcal{F})$$

into a morphism of chain complexes $C^\bullet(\mathcal{U},\mathcal{F}) \to C^\bullet(\mathcal{U}',\varphi^*\mathcal{F})$, and obtain induced maps on cohomology. We see quite directly that this construction recovers the isomorphisms on Čech cohomology which we constructed in the proof of theorem 2.7(b). The two vertical maps are the natural sheaf-to-Čech comparison maps [Har77, Lem. III.4.4], which are defined by choosing an injective resolution $\mathcal{I}^\bullet$ of $\mathcal{F}$ in $\mathcal{A}b(Z)$ and extending the identity $\mathcal{F} \to \mathcal{F}$ to a morphism of chain complexes $\mathcal{C}^\bullet(\mathcal{U},\mathcal{F}) \to \mathcal{I}^\bullet$ and passing to cohomology. Here $\mathcal{C}^\bullet(\mathcal{U},\mathcal{F})$ is the sheafy Čech complex, which is a chain complex $\mathcal{C}^\bullet(\mathcal{U},\mathcal{F})$ in $\mathcal{A}b(Z)$ with a morphism $\mathcal{F} \to \mathcal{C}^0(\mathcal{U},\mathcal{F})$ such that

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^\bullet(\mathcal{U},\mathcal{F})$$

is exact. The sheafy Čech complex is also defined to recover the ordinary one; i.e.

$$\Gamma(Z,\mathcal{C}^\bullet(\mathcal{U},\mathcal{F})) = C^\bullet(\mathcal{U},\mathcal{F}).$$

To prove that the square commutes, we follow EGA III. Just as we defined the map $C^\bullet(\mathcal{U},\mathcal{F}) \to C^\bullet(\mathcal{U}',\varphi^*\mathcal{F})$, we also have a map $\mathcal{C}^\bullet(\mathcal{U},\mathcal{F}) \to \mathcal{C}^\bullet(\mathcal{U}',\varphi^{-1}\mathcal{F})$, where $\varphi^{-1}\mathcal{F}$ is the pullback of abelian sheaves. As in the proof of lemma 2.1, we take an injective resolution $\mathcal{I}^\bullet$ of $\mathcal{F}$ in $\mathcal{A}b(Z)$ and an injective resolution $\mathcal{J}^\bullet$ of $\varphi^{-1}\mathcal{F}$ in $\mathcal{A}b(Z')$. We extend the identity on $\mathcal{F}$ to a morphism of chain complexes $\mathcal{C}^\bullet(\mathcal{U},\mathcal{F}) \to \mathcal{I}^\bullet$, and we similarly extend the identity on $\varphi^{-1}\mathcal{F}$ to a morphism of chain complexes $\mathcal{C}^\bullet(\mathcal{U}', \varphi^{-1}\mathcal{F}) \to \mathcal{J}^\bullet$. Applying global sections and taking cohomology, we get unique maps forming a square

$$\begin{tikzcd} \check{H}^i(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d] & \check{H}^i(\mathcal{U}', \varphi^{-1}\mathcal{F}) \arrow[d] \\ H^i(Z, \mathcal{F}) \arrow[r] & H^i(Z', \varphi^{-1}\mathcal{F}). \end{tikzcd}$$

Moreover, for $i = 0$ we have defined these maps so that the vertical maps are identity morphisms on $\mathcal{F}(Z)$ and $\varphi^{-1}(Z')$, respectively, while the horizontal maps are both the natural map $\mathcal{F}(Z) \to \varphi^{-1}\mathcal{F}(Z')$. As in the proof of lemma 2.1, we argue that the maps we defined on cohomology from the chain complex level are uniquely determined by the maps on H^0, so the square commutes for all i.

We recall that the maps $H^i(Z, \mathcal{F}) \to H^i(Z', \varphi^*\mathcal{F})$ and $\check{H}^i(\mathcal{U}, \mathcal{F}) \to \check{H}^i(\mathcal{U}', \varphi^*\mathcal{F})$ factor through $H^i(Z', \varphi^{-1}\mathcal{F})$ and $\check{H}^i(\mathcal{U}', \varphi^{-1}\mathcal{F})$, respectively. Hence it is sufficient to establish commutativity of the square

$$\begin{tikzcd} \check{H}^i(\mathcal{U}', \varphi^{-1}\mathcal{F}) \arrow[r] \arrow[d] & \check{H}^i(\mathcal{U}', \varphi^*\mathcal{F}) \arrow[d] \\ H^i(Z', \varphi^{-1}\mathcal{F}) \arrow[r] & H^i(Z', \varphi^*\mathcal{F}). \end{tikzcd}$$

But these are all cohomology on Z', so we see that this commutativity is precisely the naturality of the Čech-to-sheaf comparison morphism.

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